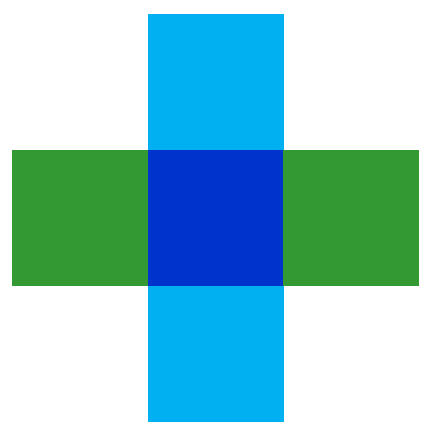




Tumor risk relapse in melanoma through a multi-modal data classification algorithm that utilises sparse clinical records and histological stains of tumor samples

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Clinical Motivation

- The Challenge: Predicting melanoma relapse from primary diagnostics is crucial for clinicians to make accurate decisions regarding tumor monitoring and adjuvant therapy.
- The Data Problem: While combining clinical records, histopathological imaging, and sequencing data improves predictions, real-world data is often sparse and poorly annotated.
- The Goal: To develop a cost-effective, efficient, and highly accurate multimodal clinical decision support tool that handles sparse real-world datasets.

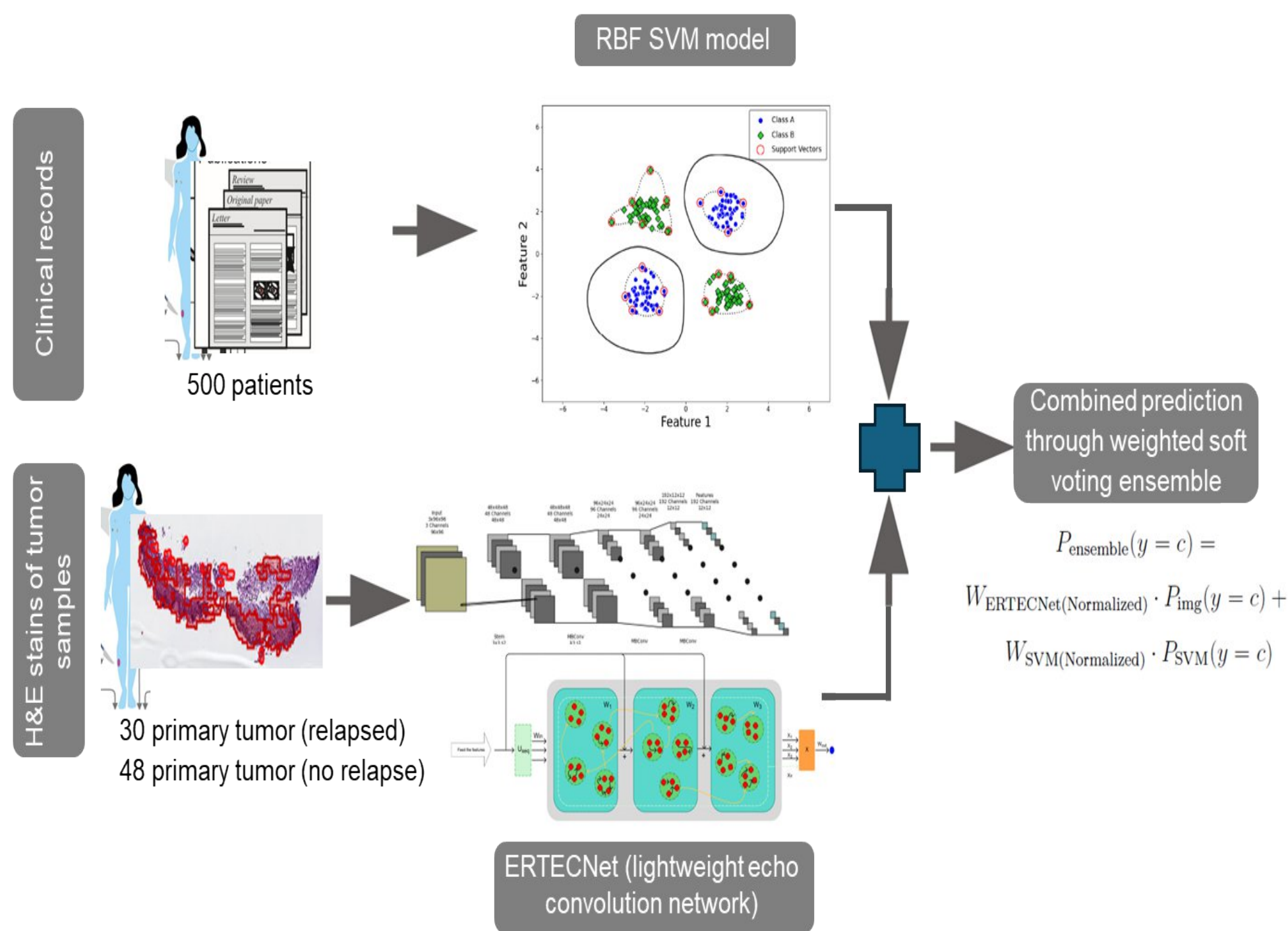
Patient Cohort & Data Curation

- Clinical Data: Curated clinical records from a large cohort of ~400 melanoma patients (177 relapse and 206 with no relapse) at the Uniklinikum Erlangen
- Histological Data: Generated and processed H&E stains from primary tumor samples for approximately 100 of these patients (including 30 with relapse and 48 with no relapse).

Multimodal Pipeline

The proposed pipeline integrates two distinct machine learning architectures to evaluate patient risk using different data modalities:

- Clinical Data Modality: A non-linear Radial Basis Function Support Vector Machine (RBF SVM) evaluates routine clinical variables to provide an initial prediction of metastatic risk (92.21% Accuracy | 98.2% AUC).
- Histopathological Image Modality: Enhanced Rapid Training Echo Convolution Network (ERTECNet), a lightweight model with fewer than 600 thousand parameters, processes Hematoxylin and Eosin (H&E) stained images.
- Ensemble Integration: The predictions from both the SVM and ERTECNet are combined using a weighted soft-voting procedure to yield a final probability score.
- The combined framework classified correctly all of the cases in a limited 8-patient validation cohort where both clinical records and H&E images were available



Patient	$P_{ensemble}(\text{Non-Metastatic})$	$P_{ensemble}(\text{Metastatic})$	Predicted Label ($\hat{y}$)	Actual Label (y)
MM10175	20.68%	79.32%	Metastatic	Metastatic
MM10283	22.74%	78.26%	Metastatic	Metastatic
MM10163	18.45%	81.55%	Metastatic	Metastatic
MM20075	17.13%	82.87%	Metastatic	Metastatic
MM10574	18.86%	81.14%	Metastatic	Metastatic
MM11206	16.75%	83.25%	Metastatic	Metastatic
MM00524	83.34%	16.66%	Non-metastatic	Non-metastatic
MM10365	68.72%	31.28%	Non-metastatic	Non-metastatic

ERTECNet: Overall Architecture

- The Concept: ERTECNet combines a lightweight CNN feature extractor with a Deep Echo State Network (DESN) [Zho*25].
- The Backbone: The feature extractor is inspired by MobileNetV3 [Qia*21] and EfficientNet [Tan*19] design principles, extracts the features from the visual data before it reaches the classification layer.
- The Mechanism: Instead of a standard fully connected classification layer, the feature extractor routes spatial features into a deep reservoir of sparsely connected neurons.

ERTECNet: The Secret to Rapid Training

- ERTECNet uses a hybrid training approach that separates how the CNN and the classification layer learn.
- Gradient Optimization: The CNN feature extractor is trained using standard backpropagation.
- Frozen Reservoir & Analytical Solving: While gradients still flow through the Deep Echo State Network to train the CNN, the reservoir's internal weights are completely frozen (fixed). The final output weights are then solved analytically via ridge regression instead of gradient descent.

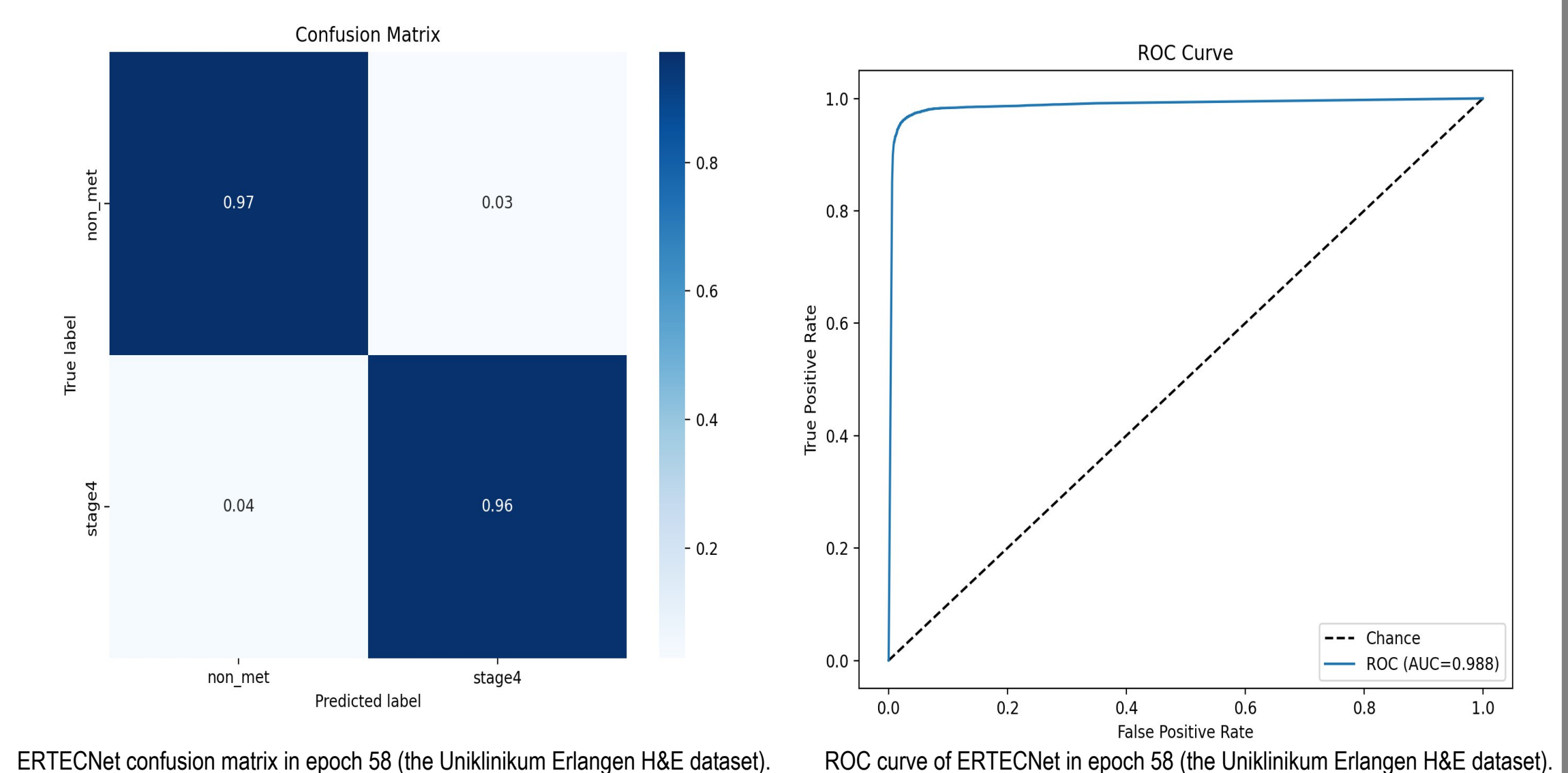
ERTECNet: Performance

- Parameters: While standard models like AlexNet [Kri*17, Zho*25] require tens of millions of parameters (e.g., ~61 million), ERTECNet operates with fewer than 600 thousand.
- Training Time for MNIST Dataset:

Hardware Environment	Model	Training Time
Server GPU (RTX A6000)	ERTECNet	7 minutes
Laptop GPU (RTX 2060)	ERTECNet	21 minutes
Laptop GPU (RTX 4060) [Zho*25]	AlexNet	230 minutes

- Despite its drastically reduced size and training time, ERTECNet remains highly competitive with heavily parameterized architectures across both baseline benchmarks and real-world medical data:

Dataset	ERTECNet Performance	Baseline Comparison
MNIST	99.67% Acc	AlexNet: 99.08% Acc [Zho*25]
K-PCAM [Kag18]	97.29% Acc 98.47% AUC	ResNet-50: 98.54% Acc 98.52% AUC [He+15,Büt+23]
Uniklinikum Erlangen H&E	96.79% Acc 98.75% AUC	(No direct comparative model)



ERTECNet confusion matrix in epoch 58 (the Uniklinikum Erlangen H&E dataset).

ROC curve of ERTECNet in epoch 58 (the Uniklinikum Erlangen H&E dataset).

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